

Example Candidate Response – Question 1, High

- 1 (a) Fig. 1.1 shows the human heart and the main blood vessels. The functions of the parts of the heart and some of the blood vessels are given in Table 1.1.

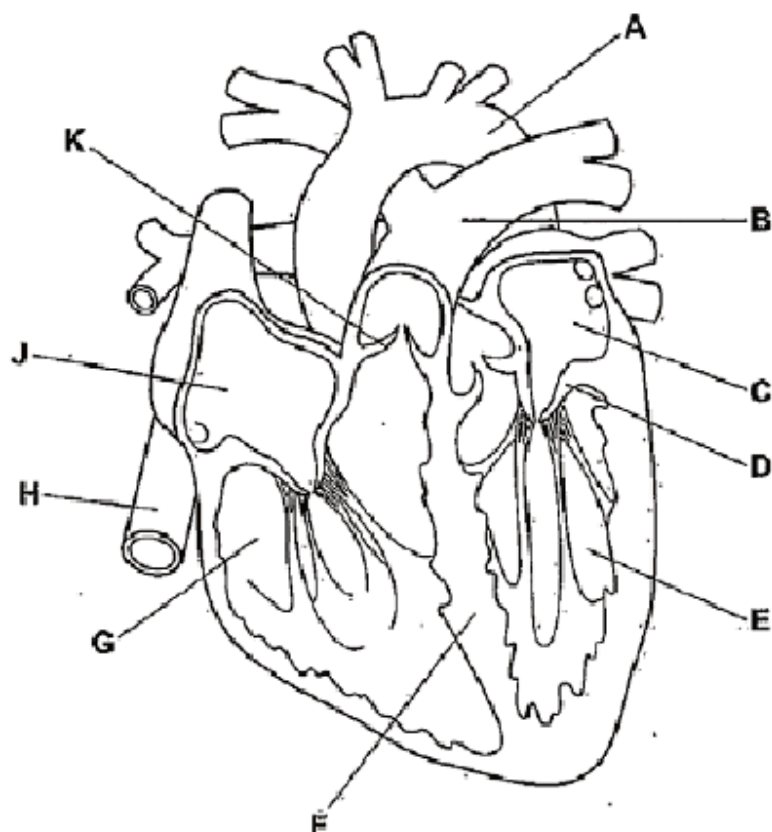


Fig. 1.1

Complete Table 1.1.

One row has been done for you.

Table 1.1

function	letter on Fig. 1.1	name
structure that separates oxygenated and deoxygenated blood	F	septum
structure that prevents backflow of blood from ventricle to atrium	D	atrioventricular atrioventricular valve
blood vessel that carries oxygenated blood	A	aorta
blood vessel that carries deoxygenated blood	H, B	vena cava, pulmonary artery

- 3 Catalase is an enzyme that breaks down hydrogen peroxide inside cells. Red blood cells contain catalase.

Some dogs have an inherited condition in which catalase is not produced. This condition is known as acatalasia and it is caused by a mutation in the gene for catalase.

- (a) Define the terms *gene* and *gene mutation*.

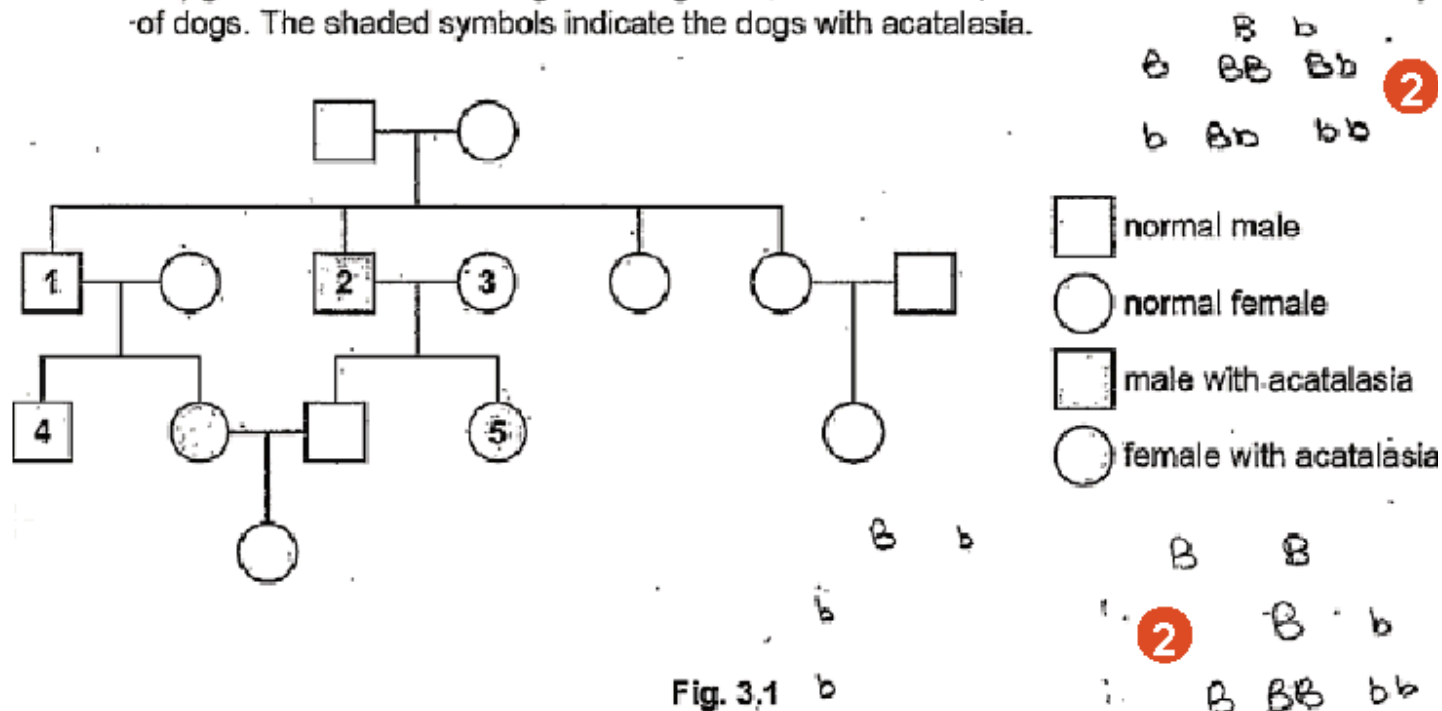
gene a length of DNA that codes for a protein.

gene mutation a change in base sequence of DNA.

- (b) A geneticist was asked to investigate the inheritance of acatalasia in dogs.

The normal allele is represented by B and the mutant allele is represented by b .

The geneticist made the diagram in Fig. 3.1 to show the inheritance of acatalasia in a family of dogs. The shaded symbols indicate the dogs with acatalasia.



- (i) State the genotypes of the dogs identified as 1, 2 and 3 in Fig. 3.1.

1 bb 3 Bb

2 bb

3 Bb

2 The nervous system coordinates the responses of animals to changes in their environment.

(a) Fig. 2.1 shows the arrangement of the nervous system in a mammal.

Complete Fig. 2.1 by writing the names of the missing parts of the mammalian nervous system in the boxes.

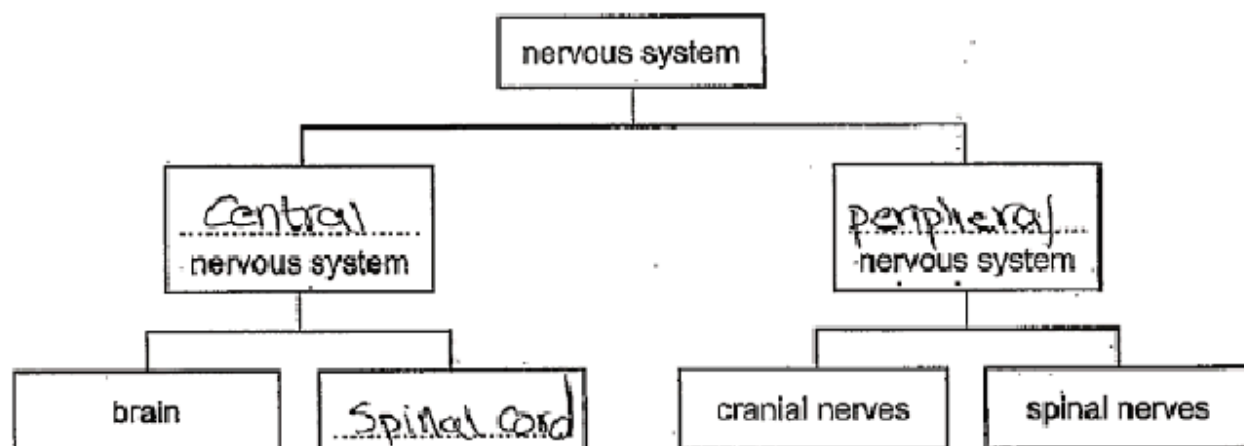


Fig. 2.1

[3]

(b) Fig. 2.2 is a flow chart that shows how an involuntary action is controlled.

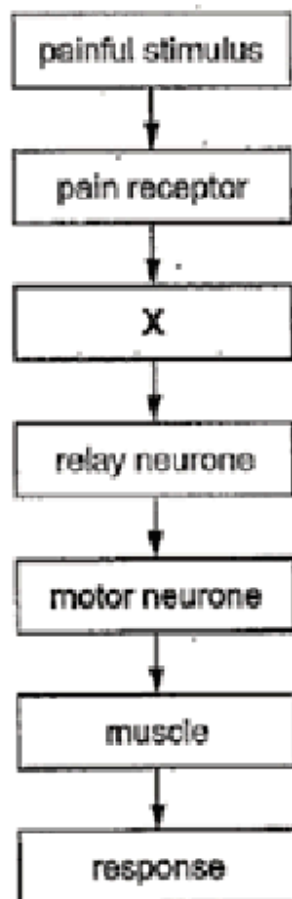


Fig. 2.2

4 *Rhabdostyla* is a single-celled organism that has no cell wall and no chlorophyll.

(a) Gases are exchanged across the cell membrane of *Rhabdostyla*.

Name:

the gas produced by *Rhabdostyla* CO_2

the process that produces the gas respiration

the method of removal of the gas ~~diffusion~~ diffusion

[3]

Rhabdostyla lives in freshwater habitats, such as ponds, lakes and rivers.

Freshwater has a very low concentration of solutes.

Rhabdostyla has a contractile vacuole that fills with water and empties at intervals as shown in Fig. 4.1. The contractile vacuole removes excess water.

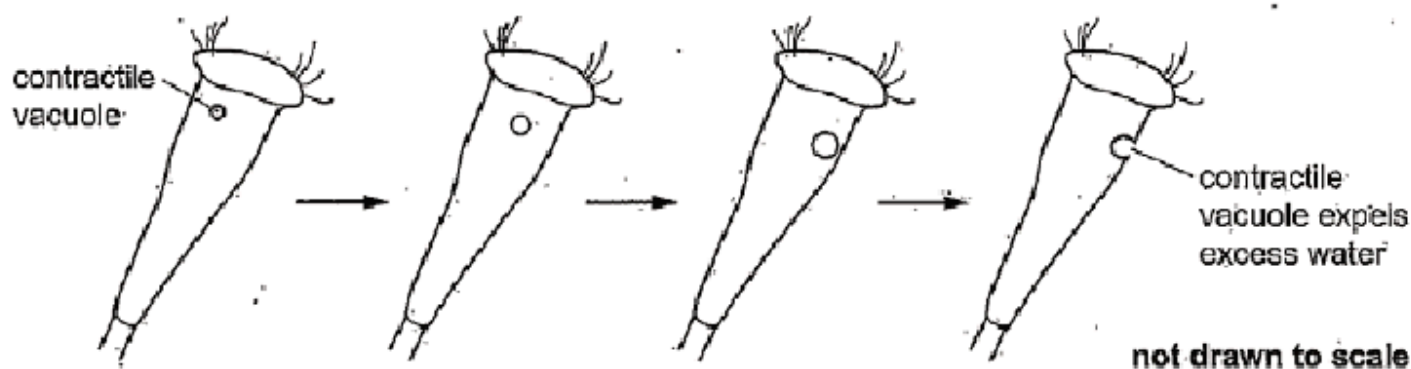


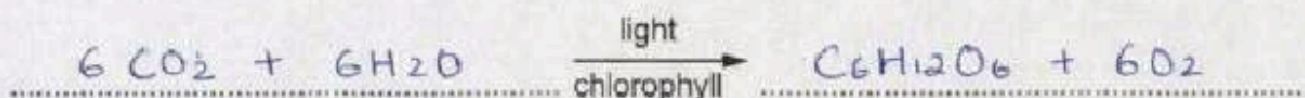
Fig. 4.1

(b) Explain, using the term **water potential**, why *Rhabdostyla* needs to remove excess water.

If *Rhabdostyla* did not remove excess it would get filled up with water until it burst as it has no cell wall to hold its shape.

as it has no cell wall to hold its shape.
It would fill up as the ^{freshwater} ~~water~~ has a high
water potential and Rhabdostyla has a
low water potential so water moves down the ~~water~~ water
potential gradient through a partially permeable
membrane into the cell by osmosis. [3]

6 (a) State the balanced chemical equation for photosynthesis.



[2]

A student investigated the effect of different wavelengths of light on the rate of photosynthesis of the water plant, *Cabomba*.

The student used the apparatus shown in Fig. 6.1.

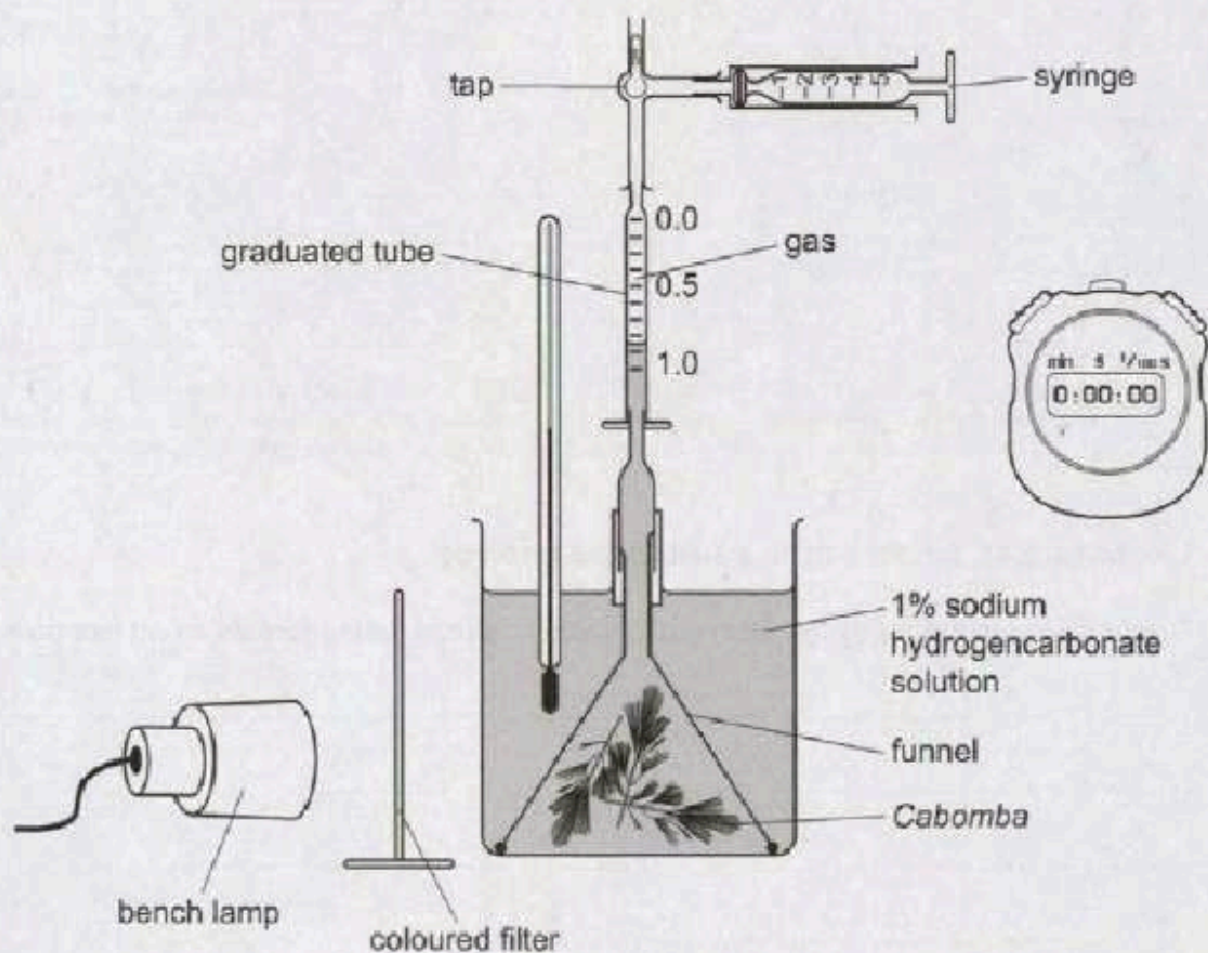


Fig. 6.1

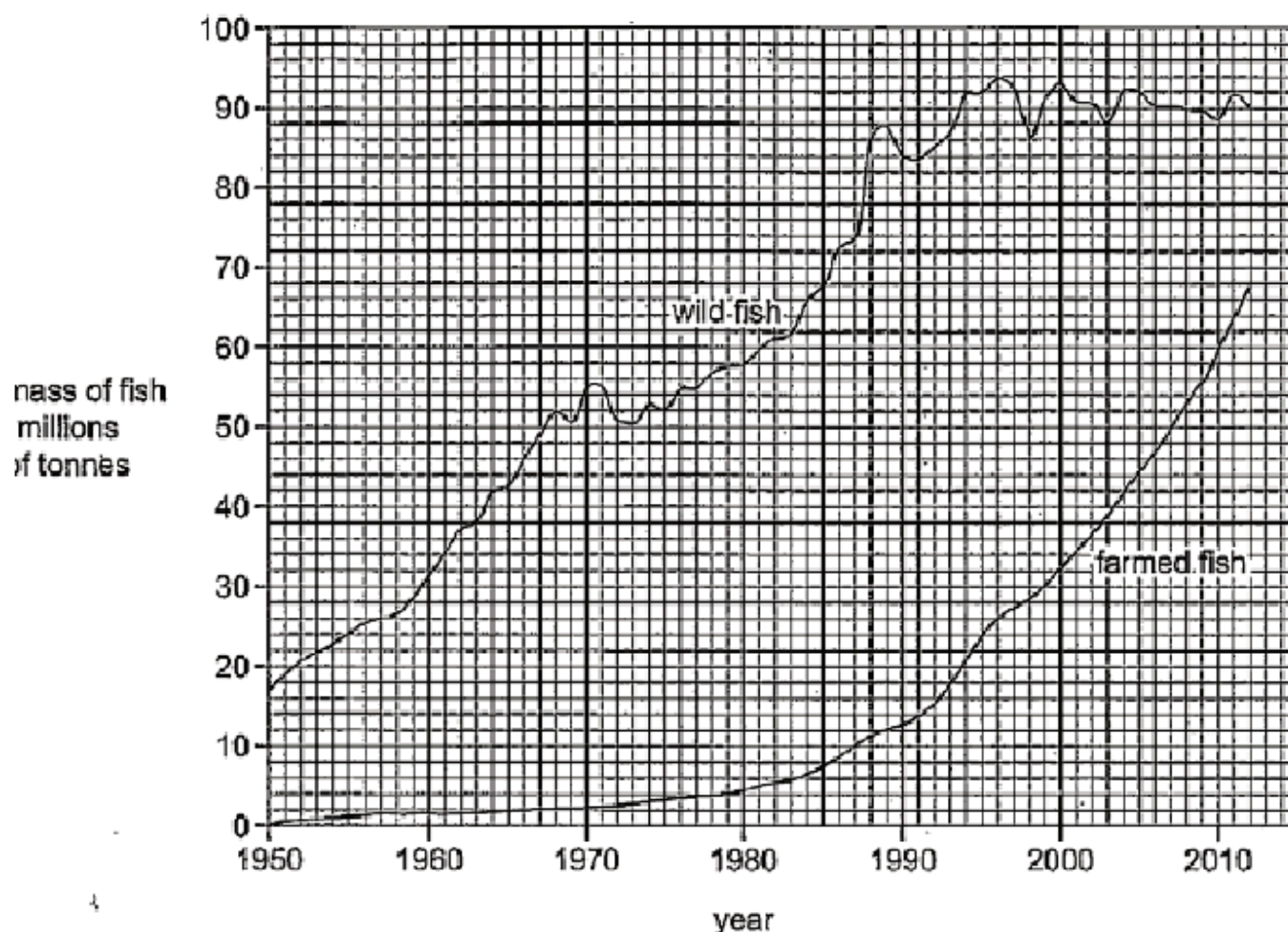


Fig. 5.1

(b) Describe the changes in the mass of **wild fish** caught between 1950 and 2012.

You will gain credit if you use data from Fig. 5.1.

There has been an overall increase in mass between 1950 and 2012. From 17 million tonnes to 90 million tonnes. It increased steeply between 1950 and 1995 and then remained fairly constant around 90 million tonnes. Greatest mass was in 1996. There were small fluctuations throughout ~~1950~~ 1950-2012.

[3]